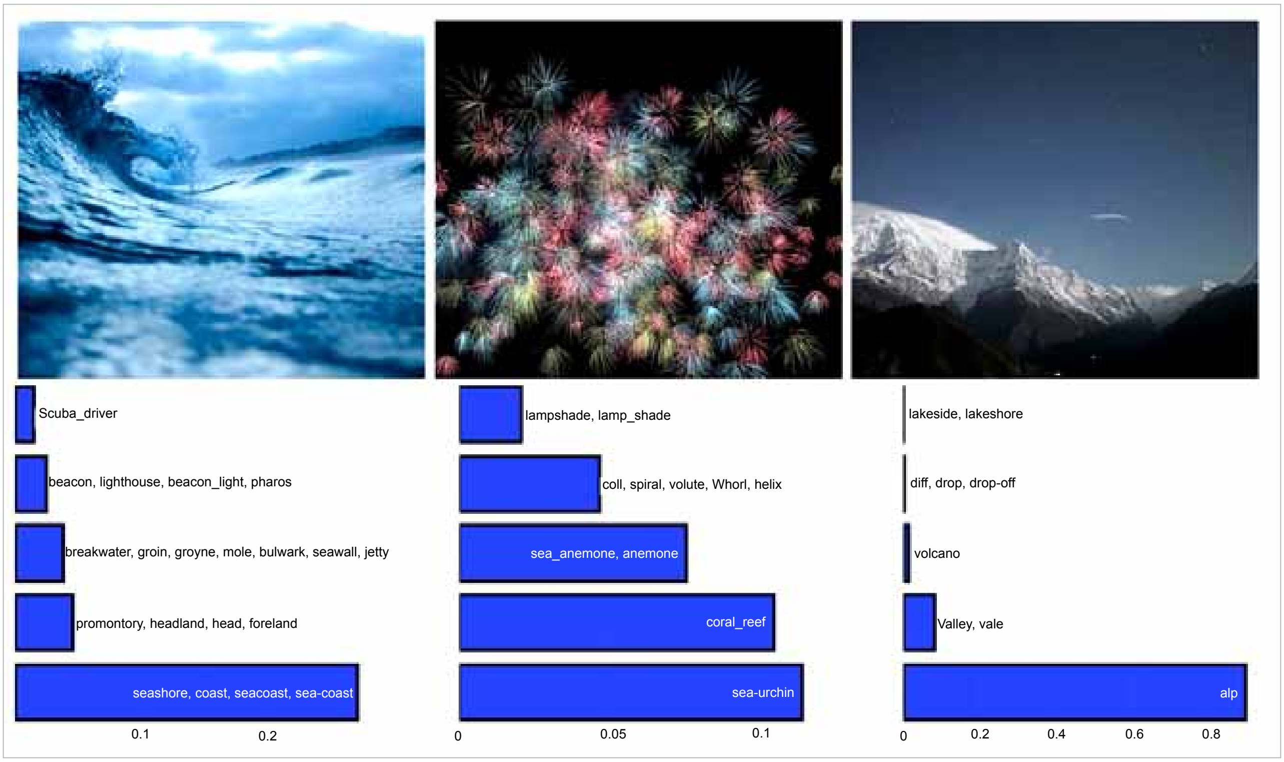

Figure 4: Image classification output of vision transformers

Limitations of DL models

Even though advances are being continuously made, DL based models presently find limited applications. Their deployment flexibility is also very low due to the following reasons:

- Huge training data available for making accurate decisions
- The device possesses high-computing power (i.e., CPU, GPU, TPU, etc)
- Uncertainty about a positive feature-engineering outcome (i.e., selecting the most suitable feature yielding the desired outcome), especially in the case of unstructured media (audio, text, images)
- Deployment restricted to high-performance devices (unsuitable for embedded, microcontrollers)
- Less or no domain expertise available

Due to their widespread applications and high effectiveness, these deep learning models are being researched and improved continuously. The focus is on improvising the optimisation and training techniques to improve speed, and make training on large data sets more efficient. Efforts are also being made to enhance the decision-making process of such models and the robustness of the system.

Here are some new and promising DL models.

Vision transformer (ViT)

Convolution struggles to relate spatially distant concepts. Each convolution filter is bound to operate only on a small neighbourhood of pixels. Relating concepts spatially apart is often vital, but CNNs struggle to do so. Increasing the kernel size or adding more depth to the CNN mitigates the problem, but it adds model and computational complexity without explicitly solving the issue. Vision transformers address this challenge by using a self-attention mechanism that operates at a 'global' scale. It is a deep learning model that applies the transformer architecture to image recognition tasks.

The model takes an input image, splits it into smaller patches, and flattens them into sequences. Then it produces lower-dimensional linear embeddings from the flattened patches and adds positional embedding. Next, the sequence is fed as an input to a standard transformer encoder to pretrain the model with image labels (fully supervised on a huge data set). Finally, it is fine-tuned on the downstream data set for image classification.

ViT is seen as an alternative to CNN in tasks that require capturing the global relationship and understanding the context of image patches. Figure 4 illustrates the image classification output of vision transformers.

So how are image classification models used for video analysis? Well, videos are just sequences of images. We make use of sequence models along with the image model to analyse videos.

While image models like ViT and CLIP are primarily designed to analyse individual images, they may